POWDER PRESSING

- The term ‘**compaction**’ is used to describe consolidation of powder particles without the application of heat.
- The main purpose of compaction is to form metal powder compacts of desired shape with sufficient strength to withstand ejection from the tools and subsequent handling up to the completion of sintering without breakage or damage.
- Compaction is followed by sintering to make the finished parts.

- In other cases, such as cemented carbides, superior properties can be achieved through hot pressing of the component powders rather than cold pressing followed by sintering.
- **Such high-temperature consolidation of powders cannot be strictly defined as compaction as simultaneous sintering is also involved.**

Powder Shaping and Compaction

- Die compaction at high pressure is an important process, where densification and shaping occur simultaneously.
- After compaction, the density can be as high as 85%.
- In the subsequent sintering step, the particles are bonded, but densification may or may not occur.
- An alternative is to use low pressures and high-binder contents to shape the powder particles.

- The binder acts as glue to hold the particles

- Most binders are polymers, which might be mineral oil or polyethylene.
- **In shaping**, the porosity ranges from 40 to 60% (excluding the polymer).
- Therefore, sintering is performed at temperatures where sufficient densification occurs.
- Fine powders assist in densification during sintering.
- Thus, key differences between compaction and shaping are in the forming pressures, initial porosity levels, particle sizes and sintering temperatures.

- Compaction relies on high-pressures and the resulting porosity is low, while in shaping, it is reverse.
- The third alternative, apart from powder pressing and shaping is the full-density processing, which relies on the simultaneous application of high pressure and high temperature, and is usually used for high-performance components used in critical applications.
- The differences between compaction and shaping technologies have an impact on the production costs.

- **Important considerations are the component**

General categorization of shaping technologies

(i) **Binder assisted extrusion**—elongated parts fine powders, constant cross-section, relatively simple shapes.

(ii) **Injection moulding**—complex, small components, high-performance materials.

(iii) **Slip casting**—very large structures, constant wall thickness, low precision.

(iv) **Tape casting**—flat sheets, very simple shapes, fine powders.

Binders

- A binder is a **temporary vehicle** for shaping the powders into the desired shape and holding the particles in that shape before sintering.
- The binder is mixed with the powder to form the feedstock.
- Most binders are multiple component mixtures that contain a major ingredient which indicates the basic properties along with additives used to obtain the other desired properties.
- As a primary requirement for easy shaping, the binder must wet the powder surface; so various chemicals are employed to improve the

- For ferrous alloys, the most common coupling chemicals are stearates including stearic acid, zinc stearate and ethylene bi steer amide.
- Of these, stearic acid is a low cost and more popular.
- These binders is very effective in lubrication as well as in bonding and burns out of the compact with minimal contamination.
- Binders reduce the mixture viscosity, allowing more solid in the feedstock.
- Normally, low-viscosity systems are desirable since high-viscosity mixtures are difficult to shape.

- For slip casting, the mixture has a viscosity similar to that of paint,
- For injection moulding or extrusion, a higher-viscosity mixture having the consistency of toothpaste is employed.
- The other important requirements of a binder include easy removal prior to sintering with minimum contamination and little or no interaction with the powder or ambient environment so as to prevent property changes over time.
- Generally, the binders are either thermoplastics or water-based gelatine systems.

- Thermoplastics allow the mixture to flow easily

- Candle wax is a common thermoplastic binder.
- Wax- polymer mixtures are widely employed in powder-shaping technologies, because of low cost, easy processing and lack of hazards.
- A mixture of two-thirds paraffin wax and one-third polypropylene can be moulded between 130 and 150°C.
- After moulding, the wax can be extracted by slow heating, immersion in solvents or evaporation by heating in vacuum.

- A related group of binders with high-water content and polymer, forms gels on cooling or drying.
- The gels are derived from organic materials like soaps, sugars and starches.
- After evaporation of the water, the remaining gel holds the particles in place until sintering.
- These are preferred for extrusion, tape casting and slip casting.
- Gelation results in a single, large molecule that extends throughout the component.
- This massive three-dimensional network molecule

- Cellulose-type gelation binders with water or glycerine are widely used in powder extrusion.
- Similar concepts are being developed for injection moulding binders.
- The binder system, though sacrificial, is an essential constituent in powder shaping.
- Several formulations are in use and various candidate binder systems can be used with any powder or shaping technique.
- Low-cost binders are preferred owing to their transient role.

Desired Binder Characteristics

(a) A good binder

(i) must have low inherent viscosity at the shaping temperature.

(ii) must be strong and rigid after shaping.

(iii) should have low molecular weight.

(b) Powder interaction: With regard to powder interaction, a binder must

(i) wet and adhere to the powder.

(ii) be chemically passive.

(iii) be stable during mixing and shaping.

(c) Binder burnout: A binder must

(i) be noncorrosive, giving nontoxic products on decomposition.

(ii) leave no residue on burnout.

(iii) have a decomposition temperature above mixing temperature.

(iv) have a decomposition temperature below sintering temperature.

(d) Manufacturing: A binder used for shaping must

(i) be readily available, safe and low cost.

(ii) have long shelf-life, low-water absorption and no volatile components.

Typical Binder Formulations

(a) Extrusion:

(i) 56% water, 25% methyl cellulose, 13% glycerine, 6% boric acid.

(ii) 72% water, 12% hydroxypropyl, methyl cellulose, 8% glycerine,

4% aluminium polyacrylate, 4% ammonium stearate.

(iii) 65% polyethylene glycol, 30% polyvinyl butyral, 5% stearic acid.

(b) Injection moulding:

(i) 69% paraffin wax, 20% polypropylene, 10% carnauba wax, 1% stearic acid.

(ii) 75% peanut oil, 25% polyethylene.

(iii) 50% carnauba wax, 50% polyethylene.

(iv) 55% paraffin wax, 35% polyethylene, 10%

(c) Slip casting:

(i) 96% water, 4% sodium ligno-sulfonate, trace calcium nitrate.

(ii) 93% water, 4% agar, 3% glycerine.

(iii) 99% water, 1% ammonium alginate.

(iv) 97% water, 3% polyvinyl alcohol.

(d) Tape casting:

(i) 77% water, 9% polyacrylate emulsion, 9% glycerine,

3% ammonium polyacrylate, 2% ammonium hydroxide.

(ii) 80% toluene, 13% polyethylene glycol, 7% polyvinyl butyral.

(iii) 47% mineral spirits, 24% isopropanol, 8% polyvinyl butyral, 8% dibutyl phthalate,

POWDER COMPACTION METHODS

(1) Methods without application of pressure:

(i) Loose powder sintering in a mould.

(ii) Vibratory compaction.

(iii) Slip casting.

(iv) Slurry casting.

(v) Injection moulding.

Methods with applied pressure:

(i) Unidirectional pressing (cold die compaction)

(a) Single-action pressing

(b) Double-action pressing.

(c) Floating die pressing.

(ii) Isostatic pressing.

(iii) Powder rolling.

(iv) Powder extrusion.

(v) Explosive compaction.

(vi) Stepwise compaction.

Pressure less Compaction Techniques

Loose Powder Sintering

- This method is also referred to as loose powder shaping, gravity sintering or pressure less sintering.
- In this method, metal powder is poured or vibrated mechanically into the mould, which is a negative impression of the article and heated to the sintering temperature.
- This is the simplest method involving low-cost equipment.
- **However, this process is not used for the production of complex-shaped parts.**
- The main reasons are:

(a) difficulty of part removal from the mould after sintering,

(b) flow characteristics of the powder, and

(c) considerable amount of shrinkage during sintering

- This method is employed to produce very large ingots up to 2 tons, which may be further processed by forging or rolling.
- The principal technological applications for loose sintering are in porous metallic materials where the amount of porosity ranges from 40 volume percent to as high as 90 volume percent.

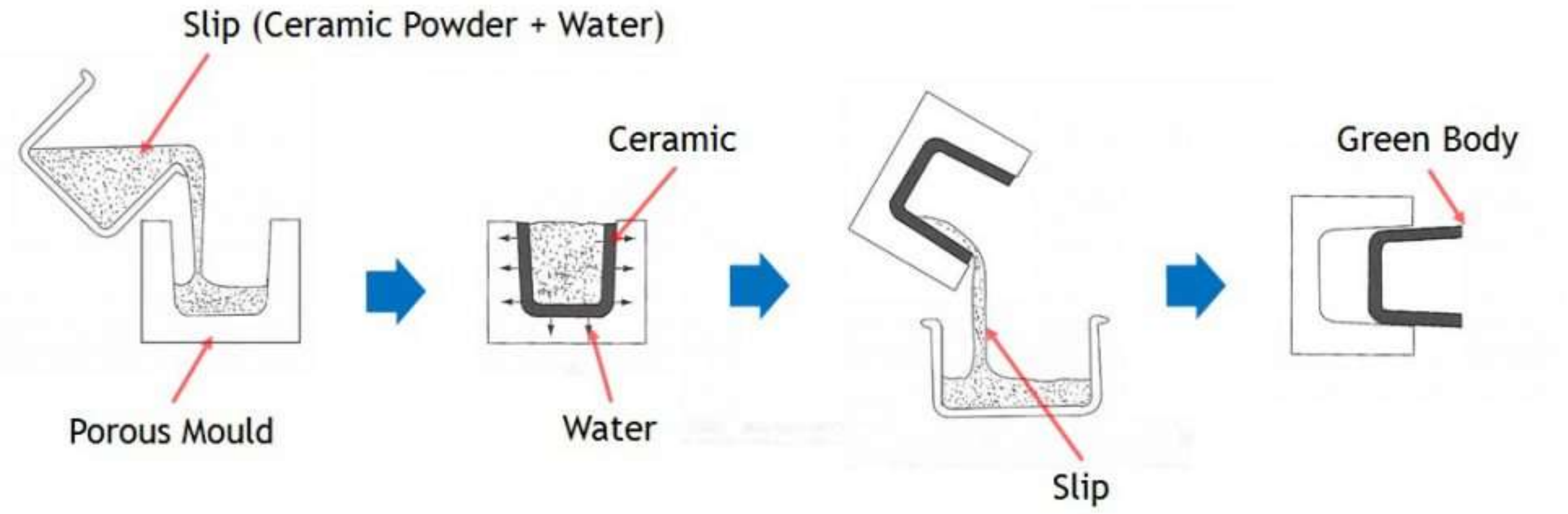
- Highly-porous filter materials made of bronze, stainless steel and monel are easily formed by this technique.
- Porous nickel membrane for use as electrodes in

Slip Casting

- Slip casting is frequently used for compacting metal and ceramic powders to make **large and complex shapes** as well as for limited production runs.
- A slip is a suspension of metal or ceramic powder (finer than 5 microns) in water or other suitable liquid which is poured into an absorbent (usually Plaster of Paris) mould, dried and subsequently sintered.
- Slips are usually made up of the **constituent powders, a dispersing agent to stabilize the powder against colloidal forces, a solvent to control the slip viscosity and facilitate casting, a binder for giving green strength** to the cast shape and plasticizers to modify the properties of

- The formation of an appropriate and a consistent slip is important for successful slip casting.
- This is usually achieved by maintaining a proper control of particle size and size distribution, addition of proper deflocculant—which prevents the settling and aggregation of powder particles and maintains the desirable viscosity of the slip.
- Mostly water is used as the suspending medium but absolute alcohol or other organic liquids may also be employed.
- Other additives that are used are alginates (ammonium and sodium salts of alginic acids) because they serve the three-fold functions of deflocculant, suspension agent and binding

Schematic representation of slip casting of ceramic powder



Advantages

- Articles can be made with shapes or in sizes that could not be possibly pressed.
- No expensive equipment is required.
- It works best with the finest possible powders, which are at the same time most-suited to sintering.
- Consequently, the finished products have excellent properties.
- The principal **disadvantage** is that the process is slow.
- It is more of an art and it has limited

- It is possible to prepare both low and high-density articles by this technique and parts such as tubes, boats, crucibles, cones, hollows, turbine blades, rocket guidance fins and other parts have been made from tungsten, molybdenum, tantalum as well as chromium powders.
- **Slip casting is also widely used for making low cost, conventional ceramic articles with excellent surface finishes like basins and water closets.**

Slurry Casting

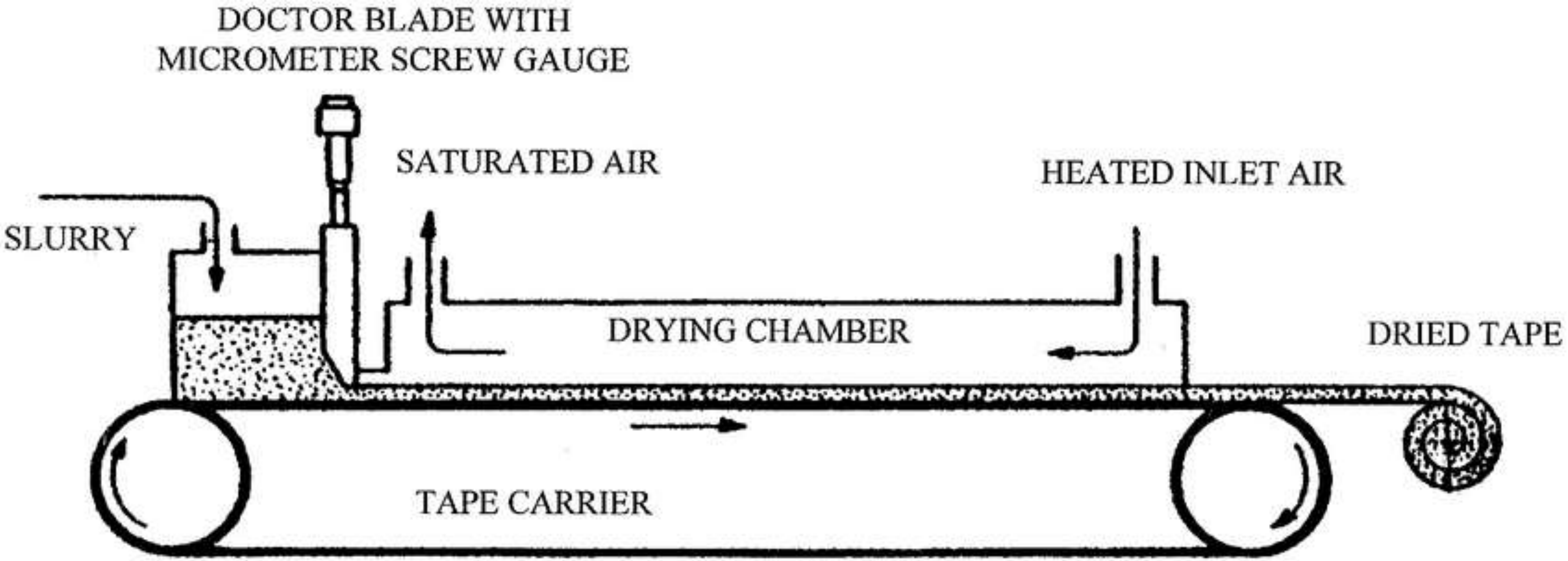
- This process is **similar to slip casting except that a slurry of metal** powders with suitable liquids, various additives and binders is poured into a mould and allowed to dry before extraction.
- The solvent may be removed either by absorption into the mould of Plaster of Paris or by slow evaporation.
- Solidifying liquid resins or self-solidifying liquids may also be used.

Tape Casting

- It is a variation of slurry casting process and is used to produce thin flat sheets.
- This process is also called as the Doctor blade process.
- The major steps are the formulation of the metal or ceramic slip (critical step), drying and firing.
- This process consists of preparing a dispersion of metal or ceramic powder in a suitable solvent with the addition of a dispersing agent to improve the dispersion of the particles.

- After a satisfactory dispersion is achieved, a binder is added and the whole mixture is fed into a reservoir.
- The slurry is then fed on to a moving carrier film from the bottom of the reservoir.

- A powder slurry layer is deposited on a carrier film (coated with lecithin to avoid sticking of slurry on to the tape) by the shearing action of a doctor blade.
- **It is important that the slurry be free of air bubbles, as it may lead to porosity in the strips, affecting the properties.**
- During sintering, the binder is burnt off first and as the temperature increases, densification



Used to produce very thin tapes, between 50 and 1,000 μm thick.

Vibratory Compaction

- In any powder agglomerate, the powder particles acquire one of the four following configurations

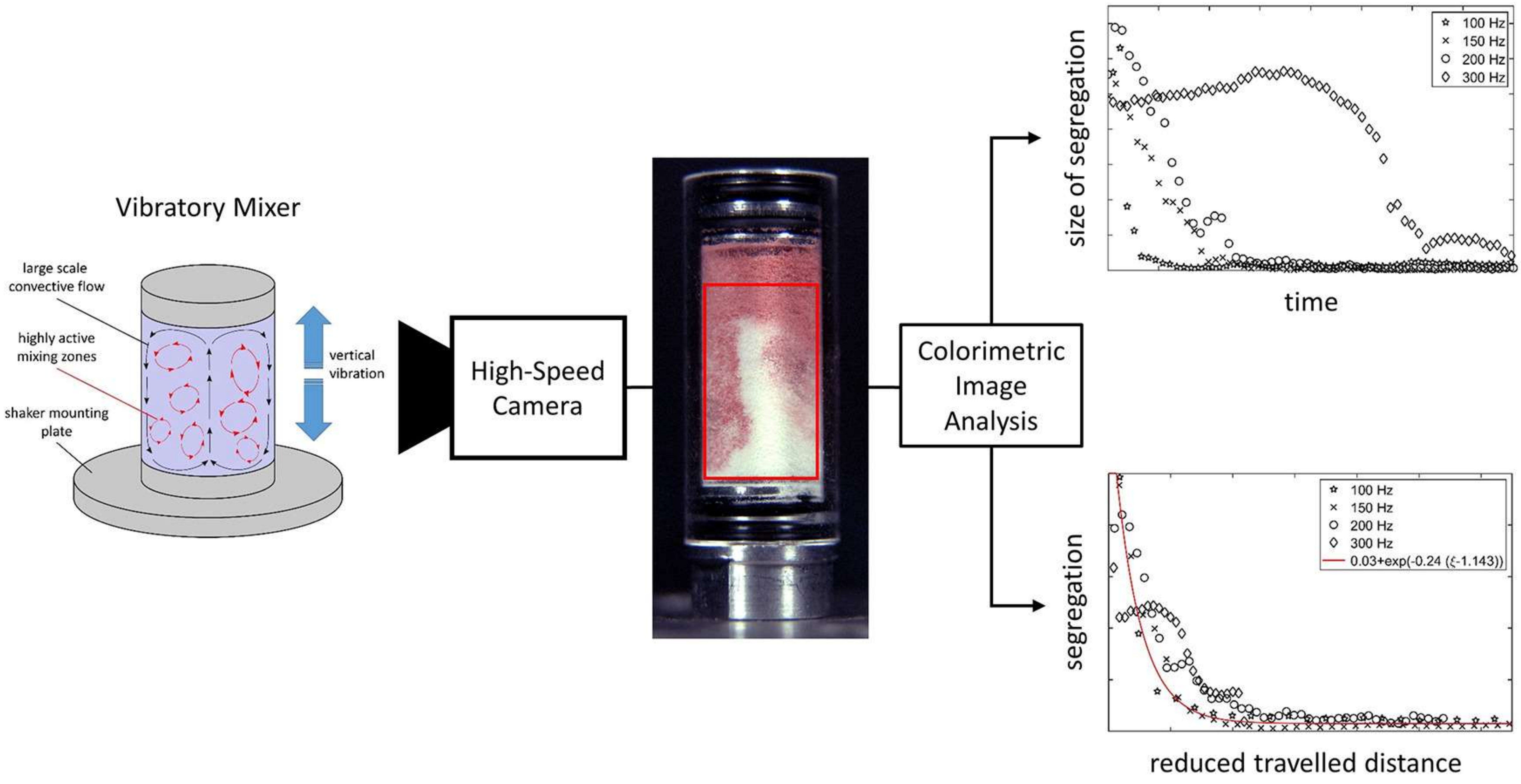
Packing	Coordination no	Porosity %
Cubic	6	47.6
Orthorhombic	8	39.5
Tetragonal	10	30.2
Rhombohedral	12	26.0

- As the size of the voids is determined by the particle size, **it is possible to fill up these voids by using particles of suitable sizes.**
- With such an addition, the size of the voids reduces further as also the total volume of voids.
- The smaller voids can also be filled with particles of still smaller size and this sequence is carried out till a high packing density of powder is achieved even before consolidation.
- An effective method of increasing the packing density of powder particles is by vibration.
- Vibratory compaction involves the use of vibrational energy for compacting loose particles into a dense mass.
- Mechanical vibration helps in settling particles in the voids present in powder agglomerate, facilitating the formation of nearly close packed powder configurations.

- This method has also an additional advantage of using only low pressures, thus, **avoiding generation of internal stresses within the compact.**

- Principle of vibratory compaction is that when a **group of particles of varying sizes are mechanically agitated**, the particles will tend to pack themselves into a close packed arrangement.

- In other words, in a mixture of powder containing coarse and fine particles, finer particles will tend to occupy the space between


Vibratory Mixer
large scale convective flow
highly active mixing zones
shaker mounting plate
vertical vibration
High-Speed Camera
Colorimetric Image Analysis
size of segregation
time
100 Hz
150 Hz
200 Hz
300 Hz
segregation
reduced travelled distance
100 Hz
150 Hz
200 Hz
300 Hz
0.03+exp(-0.24 (ξ-1.143))

- This method is generally used when

(i) powders have irregular shapes

(ii) use of plasticizers for forming is not desirable, and

(iii) the sintered density is required to be very close to theoretical density.

In vibratory compaction, **vibrational amplitude must be equal to particle size for maximum compaction**.

The number of vibration cycles in amplitude is important and not the period of vibration.

Generally, fine powders require a high frequency for effective compaction.

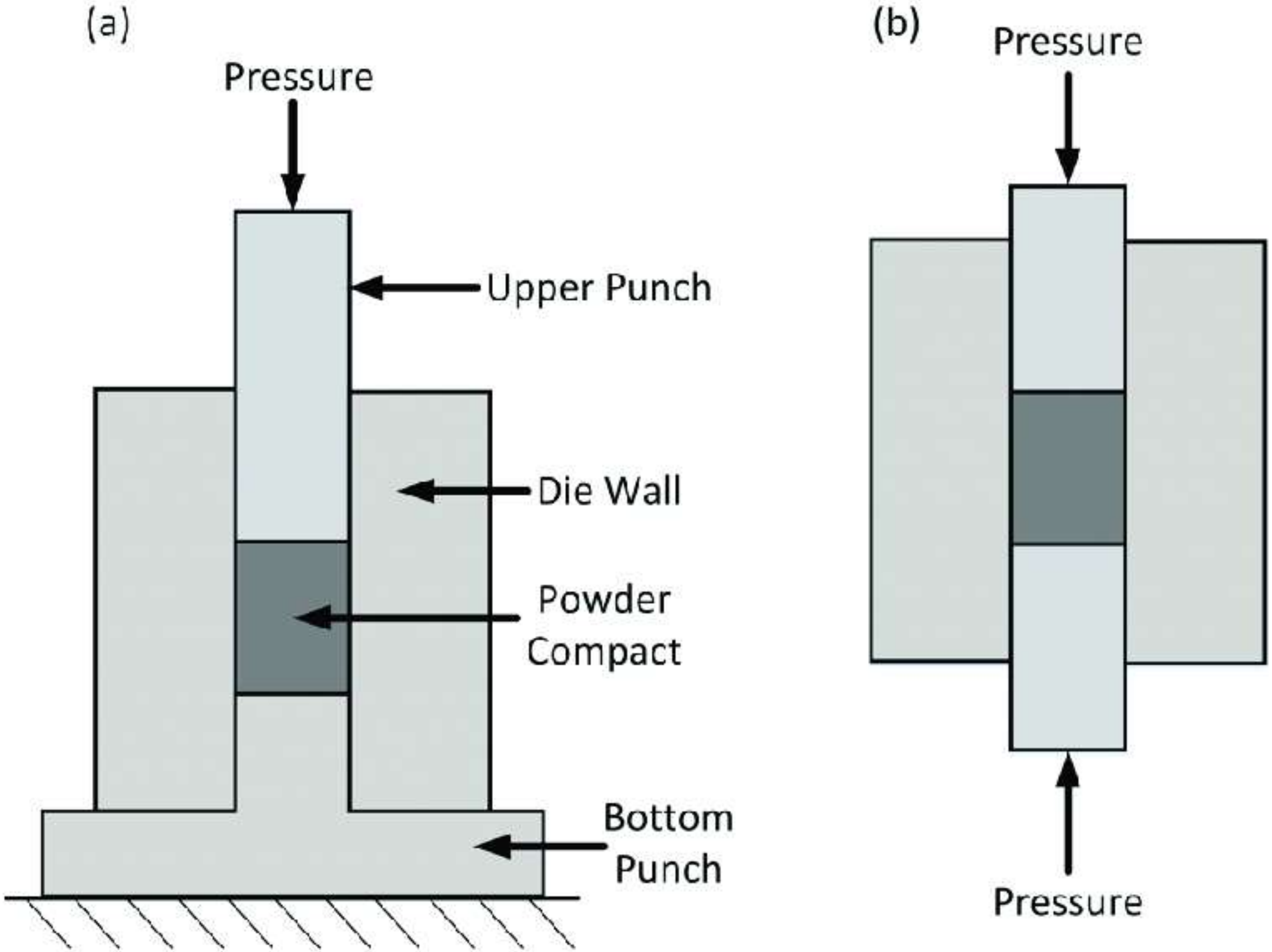
Pressure Compaction Techniques

- Pressure compaction techniques involve application of external pressure to consolidate the loose powder particles.

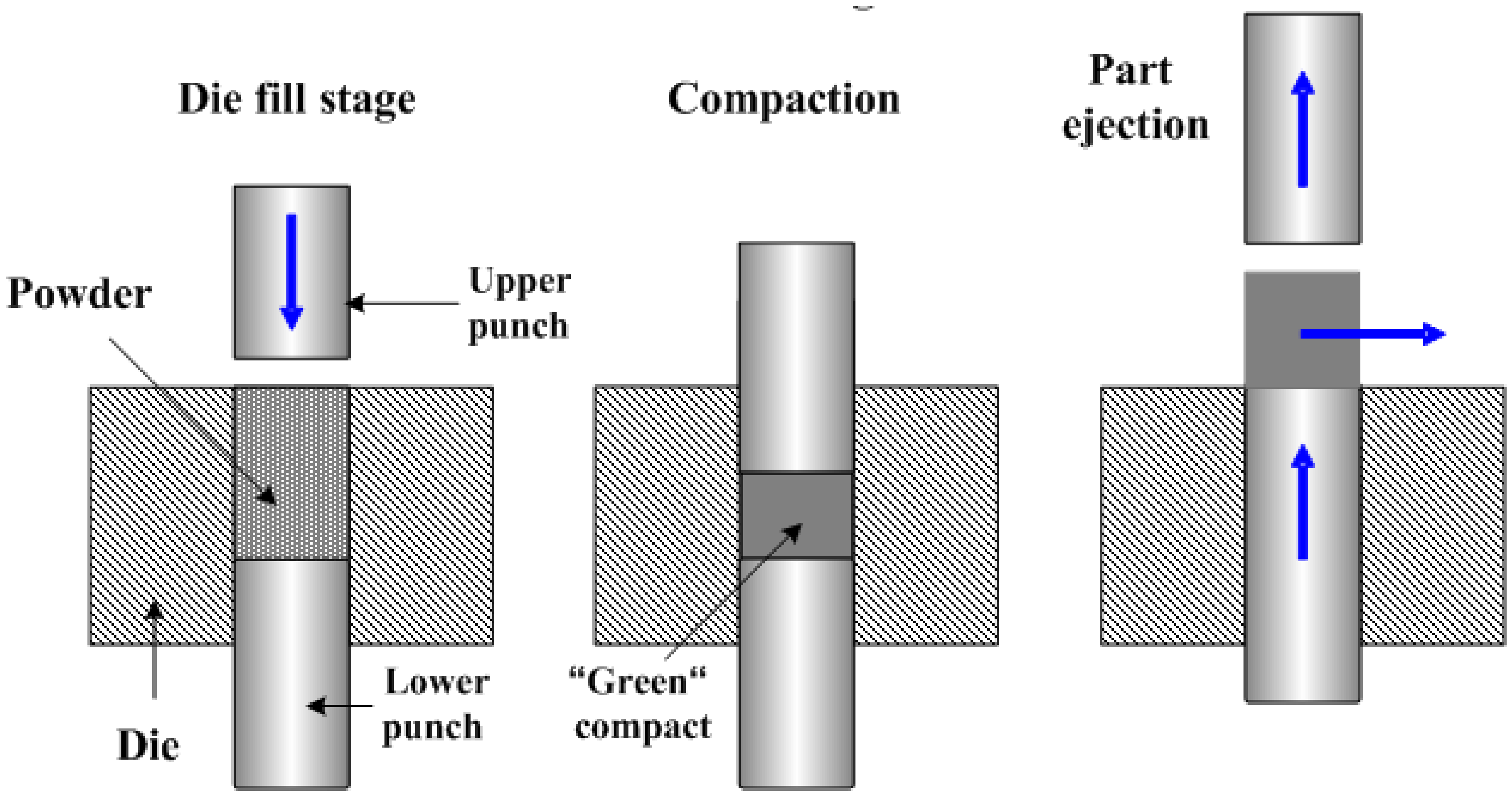
- Pressure applied may be unidirectional, bidirectional or hydrostatic in nature.

Die Compaction

- Among the various compaction methods available for pressing powders, die compaction is widely used, as it is ideal for mass production of PM parts.
- In this process, **the loose powder is shaped in a die using a mechanical or hydraulic press giving rise to densification**.
- The mechanisms of densification depend on the material (iron, copper, aluminium) and structural characteristics of the powder particles (presence of pores, or spongy powders) as well as on the geometrical factors


(a)
Pressure
Upper Punch
Die Wall
Powder
Compact
Bottom
Punch
(b)
Pressure
Pressure

Stages


Die fill stage
Compaction
Part
ejection
Powder
Upper
punch
Lower
punch
Die
"Green"
compact

Important to consider the following:

(i) Powder characteristic.

(ii) Size and complexity of the part.

(iii) Capacity of the press, and

(iv) Volume of production.

Influence of Powder Characteristics

For good cold compaction

(i) irregular-shaped particles are preferred as they give better interlocking and hence higher green strength.

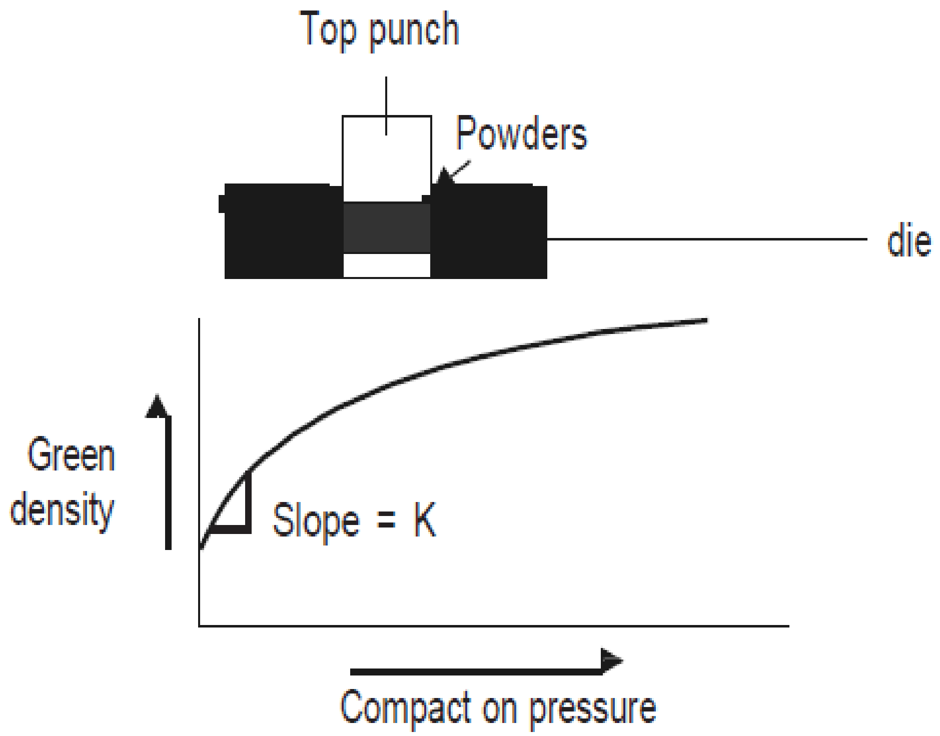
(ii) apparent density of the powders, which dictates the die fill is another important factor. Powder characteristics such as particle size, shape and density in turn influence the apparent density.

(iii) flow rate is another factor which will decide the time for filling the die. Flow rate,in turn depends on particle size, shape and

Powder Behaviour during Compaction

- Compaction involves flow of powder particles past one another during the early stages, followed by particle interaction with other particles and between die walls and punch.
- In addition, deformation of particles also takes place.
- In the case of homogeneous compaction of powders, compaction can be considered to occur in two stages: in the first stage, rapid densification occurs when pressure is applied due to particle movement and rearrangement resulting in improved packing.
- Subsequent increase in applied pressure leads to deformation of the powder particles by elastic and plastic deformation resulting in increased

- In the second stage, large increments in pressures are necessary to affect a small increase in density
- Two different regions of compaction are seen, the slope K in the first region being proportional to the inverse of yield stress.
- Above a critical press proportional to the wc powder.



- During compaction, plastic deformation causes work hardening, which reduces further deformation.
- Eventually, the particles fracture resulting in fragmentation.
- All these stages overlap.
- With increasing applied pressure, the density of the powder compact increases while the porosity decreases.
- As this initial densification is completed, increased particle-particle contact causes an increase in pressure.

- This leads to a stage where both material and geometric work hardening occur.
- At low pressures, inter particle friction predominates and the geometrical aspects of the powder agglomerate control consolidation.
- At intermediate pressures, elastic and plastic deformation of the powder occurs and the material properties become important.
- At these pressures, there is increased interlocking and cold welding of particles, leading to an increase in the green strength of the compact.
- In ductile powders, inter particle contact area increases with plastic deformation while in

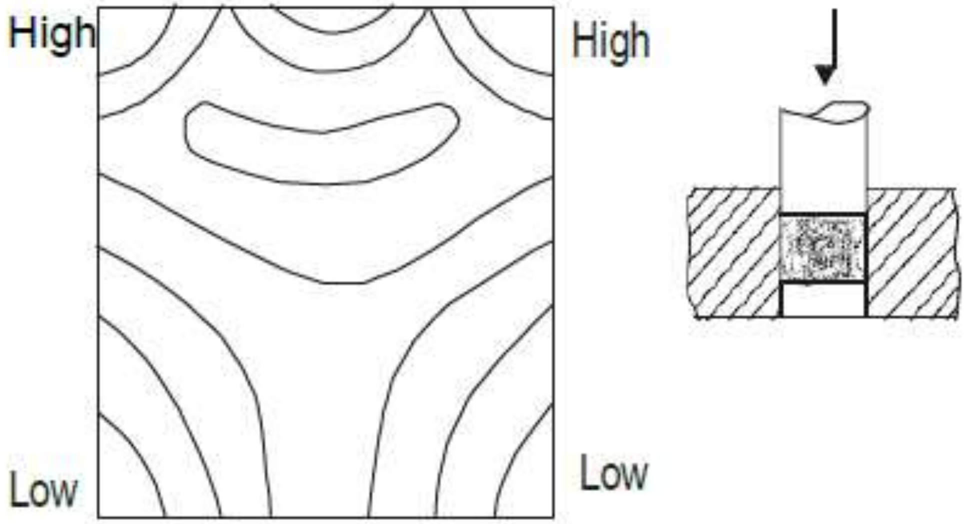
- Finally at higher pressures, the forces transmitted within the compact become more isotropic and the particles tend to come into contact with twelve nearest neighbours forming a dodecahedral configuration.
- In practice, the compaction stops when the porosity becomes largely disconnected and when further deformation of the compact becomes merely elastic.
- The green compact produced can be considered as a two-phase aggregate consisting of powder particles and porosity each having their own size and shape.

- The amount of porosity will decide the density

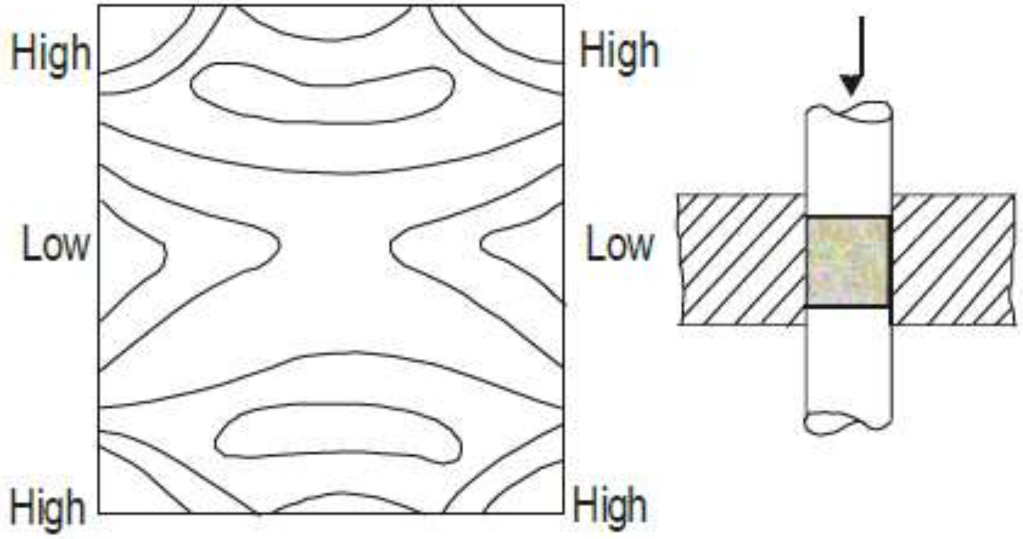
- Compaction may be done either at room temperature or at high temperature.
- High- temperature compaction gives the advantage of the modified flow behaviour of powders.
- In single-ended compaction, powders close to the punch and die walls experience much better force than those at the centre.
- The pressure difference increases with increasing length of the compact.
- This results in green density difference along the length of the compact, the density being lower at the centre than at the ends.

- The densification of the compact is not uniform, particularly at the corners.
- This is due to die wall friction and the nature of load distribution inside the die, which interfere with the compaction process.
- This green density gradient can lead to non-uniform properties in the sintered components.
- In the case of long, slender parts, low-density values are obtained near the bottom.
- This difference in green density can be reduced to a great extent by using double-ended die compaction.
- In this case, the powders experience more uniform pressure from both top and bottom, resulting in

Effect of the type of compaction on green density: (i) single-ended compaction giving variation in green density distribution, (ii) double-ended compaction giving improved green density distribution.



(i)



(ii)

- However, even with double-ended compaction, the green density variation will be significant, if the components have a high aspect (length to diameter) ratio.
- This means that long rods or tubes cannot be produced by die compaction.
- In such cases, use of advanced compaction techniques such as isostatic pressing or use of methods involving deformation and flow of material such as extrusion is made use of.

- The die compaction behaviour is affected by powder characteristics as the density increases from apparent density of the compact to theoretical density.
- In general,

(i) soft powders show initially rapid densification than hard powders

(ii) spherical powders having higher apparent densities show better densification than irregular powders during initial stages and

(iii) particle size also influences

Lubricants in Die Compaction

- In die compaction, presence of frictional forces limits the degree of densification.
- An important consequence of the die wall friction, as mentioned earlier, is a variation of green density in the compact.
- The frictional forces encountered during compaction can be minimized to a large extent by the use of lubricants, either admixed or applied to the contact surfaces.
- These lubricants also aid in the ejection of the compacts.
- Lubricants are normally organic compounds such as waxes or metallic stearates or salts and generally have low-boiling points.
- The amount of lubricant added may range from 0.5 to 2.0% by weight of the charge.

- The lubricant may be mixed with the powders and used or can be directly applied to the die walls.
- Choice of lubricant is important as it may affect the flow rate of powders or alter the die fill volume, the microstructure (formation of pores) or dimensions of the compact.
- Admixed lubricants serve to reduce the interparticle friction and aid better rearrangement and packing.
- But depending on their volume and density, they may also affect densification significantly.
- Also in the case of admixed lubricants, it is necessary to remove them before sintering, because they may cause undesirable situation such as distortion of the compact.

- The use of admixed lubricant also has an adverse

- Even a small amount (~1 wt %) of lubricant can occupy large volume (~5%) and assuming zero porosity, maximum attainable density is only 95%.
- Lubricants are, therefore, removed during a burn-off stage prior to sintering.
- Use of die wall lubricants can avoid problems caused by admixed lubricants.
- Graphite and molybdenum disulphide are common die wall lubricants.
- They are easily removed since they adhere only to the surface of the compact, but take longer production times, as it has to be applied before every operation.
- This affects the productivity adversely.

Common Lubricants Used in P/M

- Paraffin wax
- Stearic acid
- Butyl stearate
- Sodium stearate
- Aluminium stearate
- Oleic acid
- Lithium stearate
- Polyglycols
- Magnesium stearate
- Talc
- Zinc stearate
- Graphite
- boron nitride
- molybdenum disulphide

Tooling and Tool Design for Die Compaction

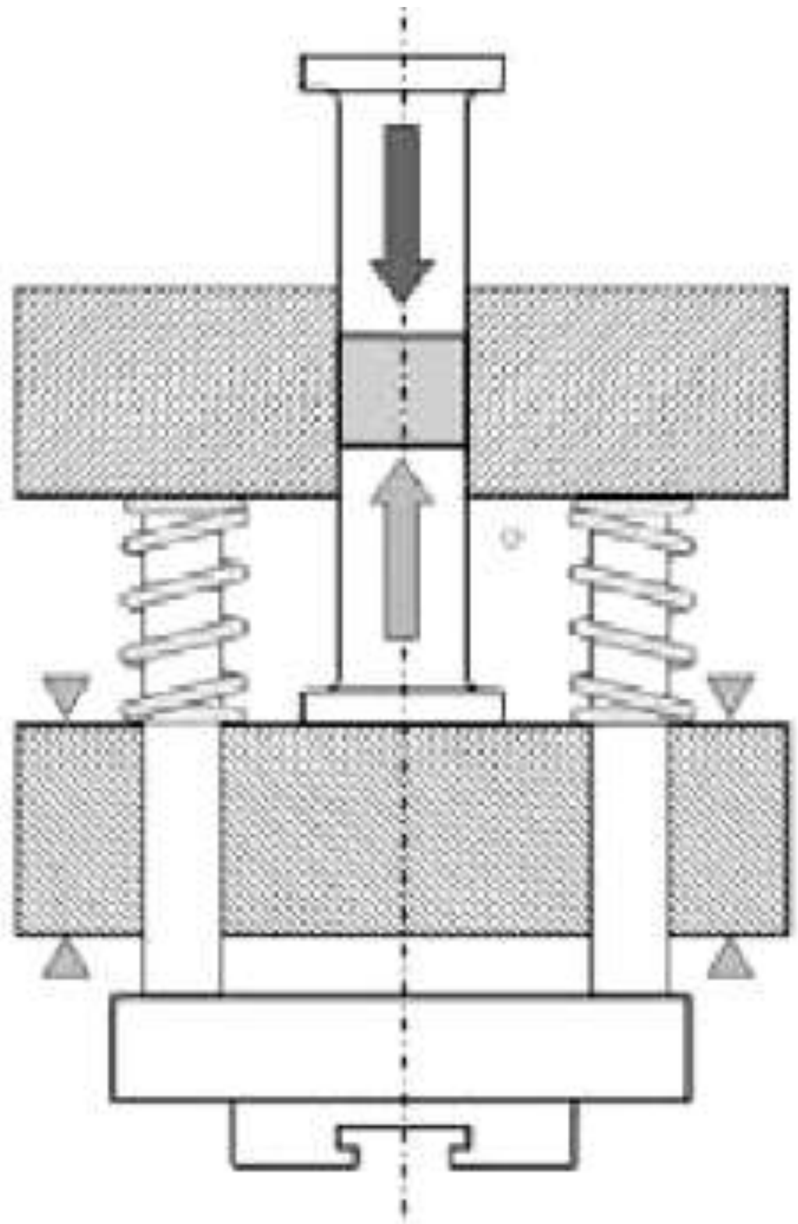
Single-action tooling systems: in which pressure is applied by the upper punch, while the die cavity and the lower punch remain stationary.

After compaction, the lower punch moves away, and the compact is ejected by the movement of the upper punch.

Simple shapes (belonging to Class I P/M parts) are produced using this type of tooling.

- **Double-action tooling:**
- In this type of tooling compaction is achieved by the simultaneous movement of both the top and bottom punches while the die and the core rod, if any, remain stationary.
- This results in better densification than single-ended compaction.
- This type of tooling is used for producing Class II, III and IV type P/M products.
- The density difference at the centre, obtained in the case of single and double-ended die compaction can be further minimized by the use of a **floating die design.**

Floating die tooling



In this design, the die is usually mounted on springs.

During compaction, the die moves downwards due to the frictional force between the powders and the upper punch.

Such a design ensures a uniform compaction.

Compaction pressure

- Limited by the load-carrying capacity, some relative motion between the die, punches and the compact.
- Floating die enables continued movement of the die after compaction, as the ejected die is withdrawn from the component, rather than its ejection by upward motion of the bottom punch.
- Withdrawal tooling is another variation of double-ended compaction used in die compaction.
- In the case of hollow parts, a core rod is usually designed as a central part of the tooling system.

- An important aspect of die compaction is the design of the dies.
- In the design of the dies, factors such as shrinkage, lubricant characteristics, die material properties, the surface finish and dimensional tolerances achievable by machining are to be taken into account.
- Tool design for compaction must address the problem of limited lateral flow as well as ease of ejection after compaction.

Powder characteristics affect tooling design

(i) Flow of powders: Powder should flow freely enough to fill the die.

Flow of powders can be improved by the use of lubricants.

(ii) Die fill: This is the amount of powder in the cavity prior to compaction.

Die fill depends on particle characteristics like shape (which affects apparent density and flow rate) and geometry of the part.

Ideal filling conditions are those which permit free flow, uniform fill density without choking or bridging and an optimum filling rate of the tooling cavity, complexity of the part such as thin sections

Equipment for Die Compaction

- Die compaction is generally done using mechanical or hydraulic presses.
- Combinations of hydraulic and mechanical or with pneumatic options are also used.
- **The presses meant for P/M production have to be much stronger because the clearances used in P/M compaction are very small.**
- **The major advantage of hydraulic presses is that the speed can be controlled: fast, medium or very low speed depending upon the need can be obtained but costlier and NOT suitable for fast production.**

Major defects in die pressing

- **Lamination cracking**:

If the air is not allowed to escape from the die during compaction, it gets entrapped and results in lamination cracking.

This cracking occurs in a direction perpendicular to the direction of pressing.

The entrapped air prevents the interlocking of particles.

This situation arises when the components are pressed too fast to permit escape of entrapped air.

- **Blowout:** This defect occurs when all the entrapped air tries to escape at the

CLASSIFICATION OF P/M PARTS

- P/M parts are classified based on their complexity.
- Metal Powder Industries Federation (MPIF) has defined a classification system, which identifies four levels of P/M components made by conventional die pressing

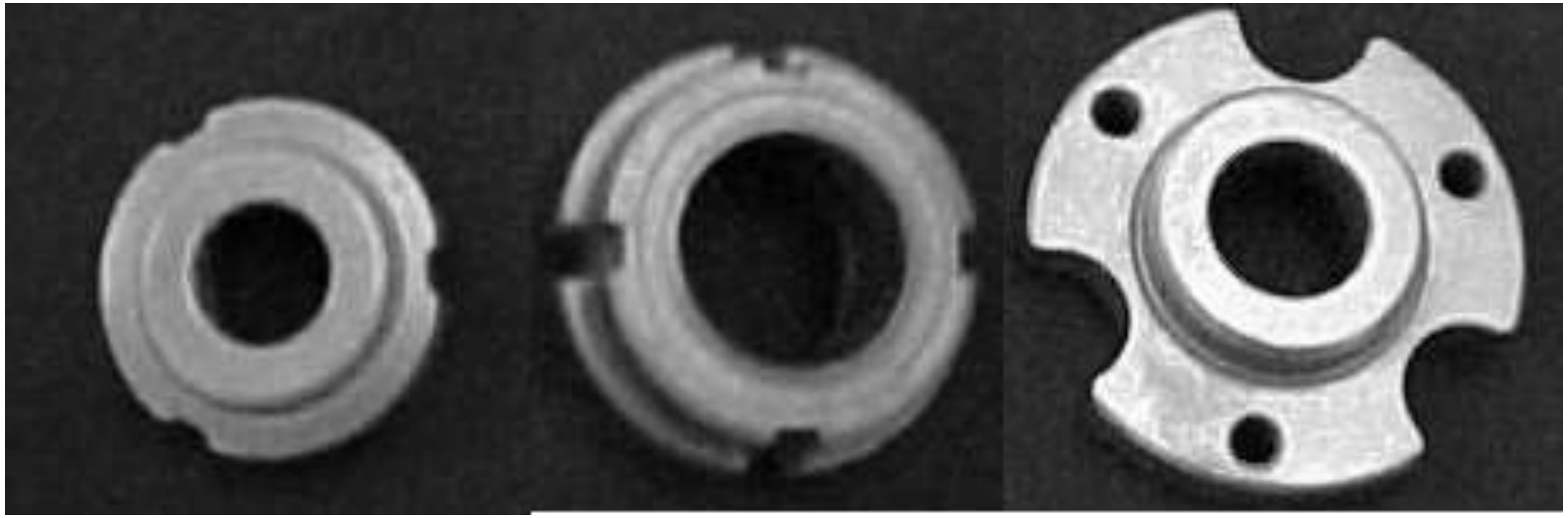
Class I: Simple and thin sections, unidirectionally-pressed with single level compaction.

Class II: Simple, but thick sections requiring pressing from two directions.

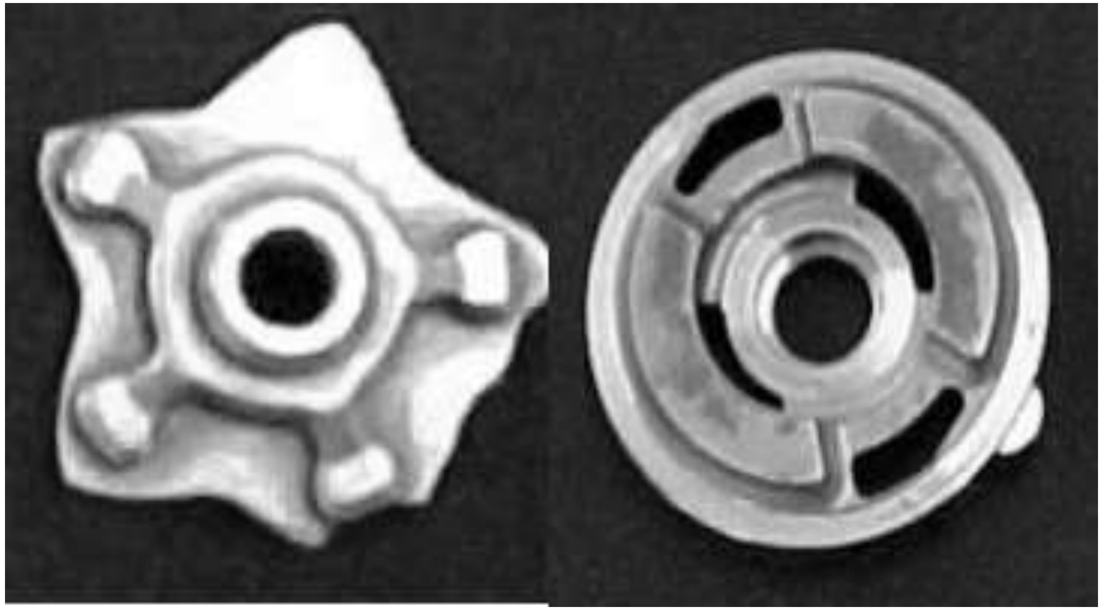
Class III: Two levels of thickness requiring pressing from two directions.

Class I parts

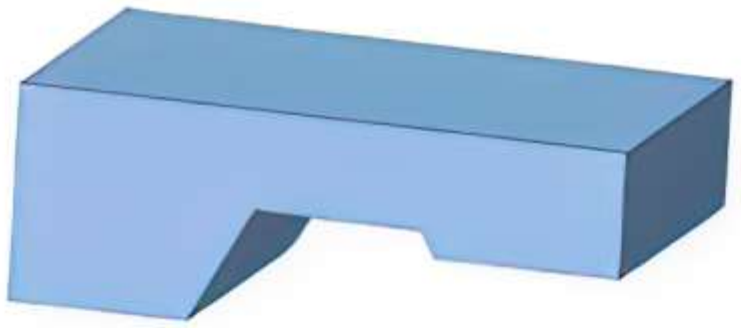
Class II parts



Class III parts



Class IV parts



COLD ISOSTATIC COMPACTION

- Conventional die compaction has a major disadvantage in that the size and shape of the parts are limited both by part geometry as well as by the capacity of the press.
- Die compaction is also not suitable for complex-shaped parts because of difficulties in die fabrication and subsequent ejection of the compact.
- These limitations are overcome in Cold Isostatic Pressing (CIP), which provides an economical and effective alternative for producing high-density compacts.
- The powder parts can be consolidated to densities up to 80 to 90% of their theoretical densities in CIP.
- The process was first developed and patented by

Process

- In Cold Isostatic Pressing, pressure is applied simultaneously and equally in all directions, using a fluid (water or oil) to an elastomeric mould filled with powder at room temperature.
- CIP is utilized to form green parts of simple or complex shapes to net or near-net shape in one simple, fast, high-pressure operation.
- Sintered cold isostatically pressed component can reach around 95 to 97% of theoretical density with enhanced mechanical properties.
- Further increase in density may be achieved by working the sintered part or by Hot Isostatic Pressing (HIP).
- Tooling is typically an inexpensive, flexible mould or container made of rubber or plastic.
- CIP is not shape or size-limited and it results in

Steps

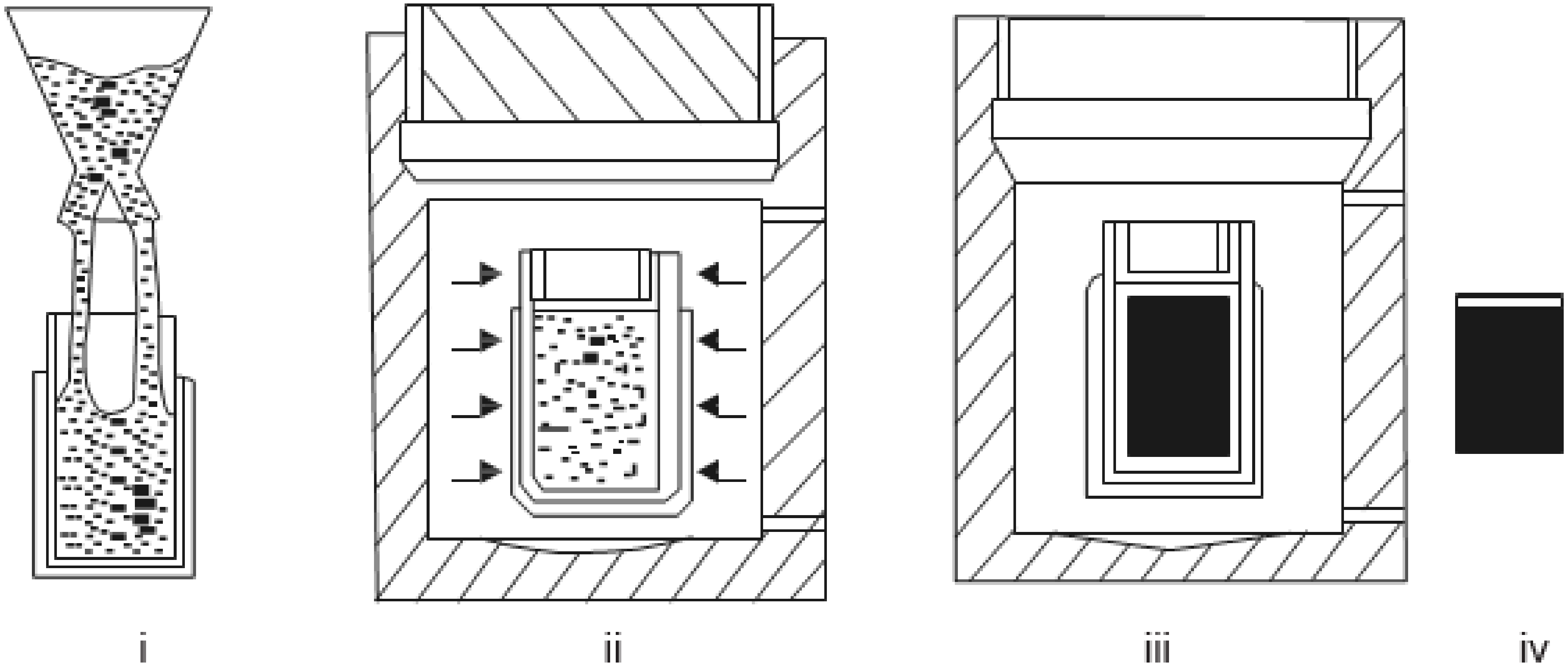
(i) Filling of the powder in a flexible capsule and evacuating it.

(ii) Placing the sealed capsule in the pressure vessel.

(iii) Application of pressure using a liquid.

(iv) Release of pressure and removal of the compact.

(i) mould filling, (ii) pressurizing the mould, (iii) depressurizing, and (iv) green part



- The powder is taken in a flexible mould-shaped like the end product.
- The mould is filled with the powders, evacuated and sealed, and subjected to hydrostatic compressive stresses inside the high-pressure vessel.
- Thus the powder is compacted with the same pressure in all directions and high and uniform density can be achieved without any lubrication.

- **Good mould filling is essential in isostatic pressing because the initial powder distribution and density influence the bag movements significantly and hence the preform shape.**
- Powder particle size, shape, density and mechanical properties influence the ability of the powder to flow into a mould (measured by flow

- Optimum filling is obtained by the use of a free-flowing powder along with **controlled vibration or tapping of the mould.**
- Vibration not only promotes good mould filling but has also the desirable effect of increasing the bulk density, thus, reducing the effective compaction ratio.
- **Materials for flexible moulds must be resilient**, exhibit uniform elastic behaviour in all directions and be unaffected by the liquid used for pressurisation.
- The materials normally used are natural and synthetic rubbers such as neoprene, urethane, butyl, nitrile and silicones.

- Design of pressure vessel must ensure complete safety during operation.
- In many cases, vessels have a **leak-before-break** design to make the operations highly safe.
- In a porous powder mass, during compaction, the **hydrostatic pressure is resolved into shear stresses**, which are large near the pores but decrease with distance away from the pores.
- This causes effective compaction during pressing.
- With increasing density (or decreasing porosity) these stresses become negligible and compaction ceases.
- Initially, the resolved shear stress serves to increase the density of the compact by particle sliding and rotation, which depends on the packing geometry and the flow properties of the powders.
- In the next stage, deformation of powder particles occur and particle characteristics such as shape play an important role during this stage.
- **Irregularly-shaped particles**, which interlock with one another and also deform during both the stages, tend to

- In the case of **spherical powders**, in spite of their higher initial packing densities, particles do not mechanically interlock with one another and hence do not easily deform.
- Higher pressures are required for their compaction.

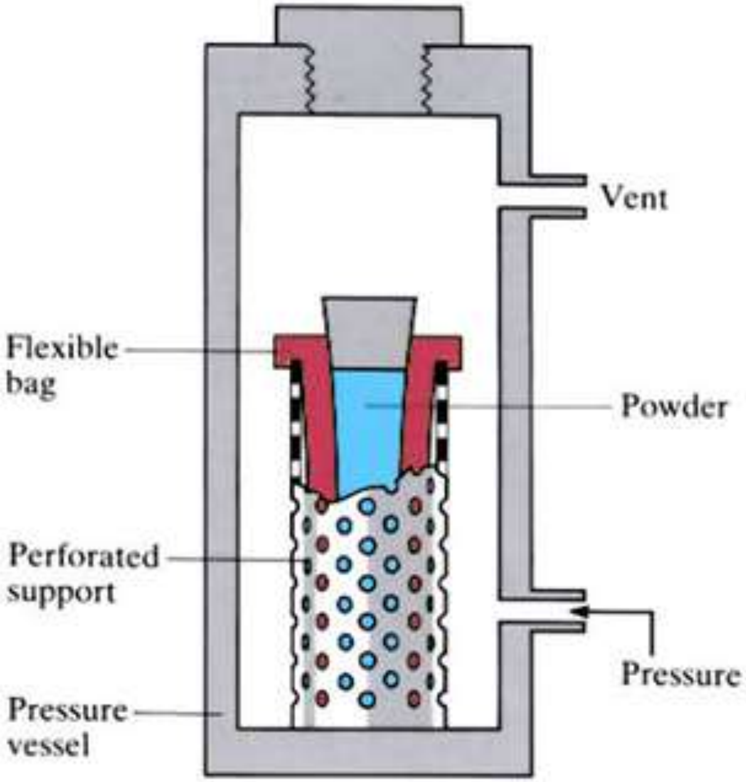
- After compaction is over, the **rate of depressurisation** of the chamber must be controlled.
- This is essential to avoid cracking of the compact due to the release of entrapped gas in the compact.
- The compact is then removed from the bag and

Types

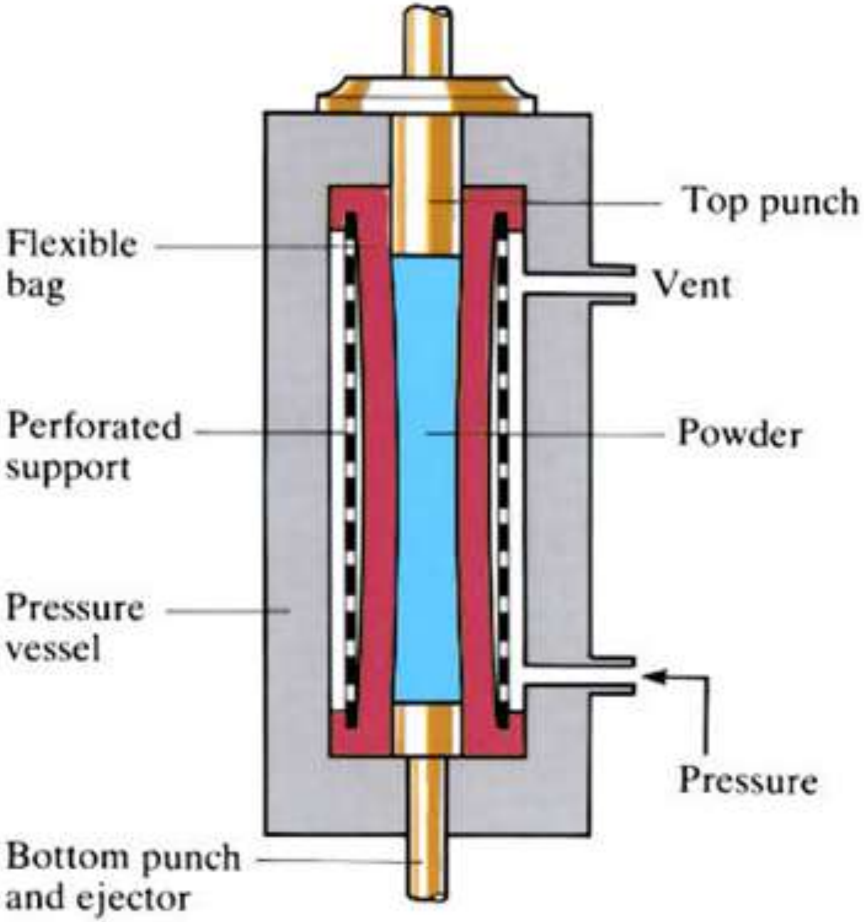
- Wet-bag
- Dry- bag processes.

- Both the processes make use of an elastomeric mould, resembling the final part, in which the loose powder is kept.

- In the wet bag process, the mould is removable, while in the dry bag process, the mould is fixed to the pressure vessel.



WET-BAG



DRY-BAG

- In the wet-bag process, the mould is directly in contact with the fluid. This reduces the productivity, since the bag has to be removed every time before refilling.
- **Tooling costs are reduced**

- In the dry-bag process, the mould is fixed inside the pressure vessel, and powders are filled in situ.
- The mould is separated from the pressure fluid by a sleeve made of an elastomer, i.e. the tooling itself is built with internal channels into which fluid is pumped.
- This is an automated process in which the powder filling, compaction, depressurisation and removal of green parts are done continuously.
- The dry bag process gives higher production rates due to the faster cycle times.
- **Higher-tooling cost.**

Advantages

- Compacts with uniform density relatively free from cracks, strains and laminations caused by residual stresses.
- Reproducible density in compacts enabling better control of dimensions after sintering.
- Higher green strength, allowing better handling and machining.
- Uniform and controlled densification of the powder
- Economy of operation even for complex and large parts.
- Low tool costs (due to the use of cheaper elastomer moulds).
- Flexibility of operations in that different shapes can be pressed simultaneously.
- Long and slender parts [with high length to diameter (l/d) ratios] can be pressed.
- Near-net shape forming technique.
- Reduction in the amount of binder necessary for

- Major disadvantages include high equipment cost and relatively lower productivity compared to conventional die pressing.

- **Defects such as necking (due to insufficient powder fill) cracks are possible with both wet and dry bag processes.**

Applications

- Used for producing semi finished (billets, preforms) as well as near- net shapes.
- Metallic filters made from bronze, brass, stainless steel, Inconel, Hastelloy, Monel, titanium, high speed tools, tungsten carbide tool bits and parts with complicated shapes as well as threaded bushings can be successfully fabricated.
- Widely used for pressing of ceramic parts such as sparks plugs and insulators.
- An electrical insulator made by CIP process is shown in Figure

- Wet bag pressing units are used for larger and complex parts, which are needed in smaller quantities.

- The process is flexible and can process various sizes and moderately complex shapes, chief limitation being the volume of the pressure vessel.

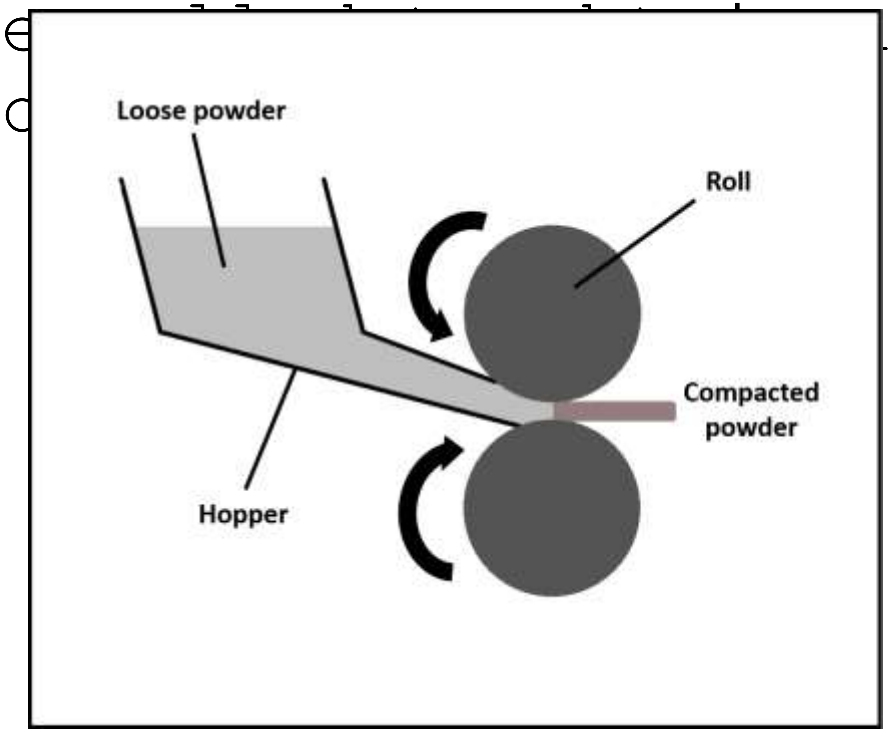
- Dry bag process is used for smaller parts such as tubes, rods and nozzles.

POWDER ROLLING

- Powder rolling was first attempted by **Henry Bessemer** in 1840s when he tried to produce flake gold by passing gold powder between two rolls.
- Instead of flakes, a continuous strip was obtained.
- Powder rolling, however, became an important technique only in the **1950s when rolling of iron powder** was carried out.
- Since then, powder rolling process has undergone several modifications and is used widely for making bimetallic and trimetallic strips as well as coinage strips.

- **Powder rolling is advantageous in that it enables**

- Powder rolling has been defined as "**the feeding of a powder between rolls to produce a coherent but brittle green strip**." This green strip is subsequently sintered and re
dense, finishe



- In powder rolling, metal powder with binders (used to improve green strength) is fed from a hopper to a set of rolls in a rolling mill to produce a continuous green strip or sheet.
- Powders can be elemental, blended or pre alloyed powder having good flow and compaction characteristics.
- Rolls are normally positioned horizontally, though they can be positioned vertically or even at an inclined angle.
- **Use of horizontal rolls and gravity feed offers flexibility in controlling powder feed and in introducing different powders simultaneously for multilayer strip production.**
- The important parameters in powder rolling are the control of powder feed, roll diameter, edge control, sintering and re-rolling. Height of the powder bed in the hopper
- In practice, the roll gap and the roll friction are adjusted to give a green density of 75 to 90% of the theoretical density.
- **For compaction to occur, the normal force acting on the powders must exceed the frictional forces.**

Difference between powder and conventional rolling

(i) the starting materials are powders and not solid blanks, requiring direct feeding as they cannot be gripped by tensile forces as in conventional rolling

(ii) during rolling, powders undergo changes not only in density, but also in volume. In conventional rolling, the volume is constant.

- **Like conventional rolling, this process is also continuous and different from other P/M compaction techniques.**

Steps in Powder Rolling

(1) Preparation of green strip
(2) Sintering
(3) Densification of sintered strip
(4) Final cold rolling and annealing

Preparation of green strip

- This step consists of feeding the powder mix between the rolls to obtain a green strip.
- Three different stages have been identified in the production of green strip, which may overlap one another:

(i) Rearrangement/restacking of particles due to slip between particles.

(ii) Elastic and plastic deformation of particles.

(iii) Predominant plastic deformation of particles.

- Physically, when the powder passes between the rolls, it encounters three zones, viz.,

(i) Free zone: The region where the powder mass is loose and there is no physical bonding between the particles.

(ii) Feed zone: The region where the powder mass gets drawn into the roll by friction.

- This results in slip between particles and causes restacking of particles leading to the formation of porosity distribution.

(iii) Compaction zone: The powders undergo flattening due to rolling and a deformed aggregate of powders with porosity is obtained.

- In powder rolling, the thickness of the green strip obtained will not be equal to the roll gap.
- This is due to the elastic flattening of rolls during rolling as well as elastic spring back of deformed powders after rolling.
- Release of entrapped air also causes expansion of the strip.
- In powder rolling, rolling is done only once, and hence proper design must be done to take care of strip thickness and density.
- Based on density and green strength requirements, it is possible to design the mill with proper roll gap.
- The change in dimension of the strip is dependent on powder characteristics, roll die and compaction speed.
- Feeding more powder increases the green density of the strip and resulting expansion due to entrapped air is less.
- Similarly, lower compaction speed gives more time for

- Powder feeding is done either under gravity or by force-feeding (using a screw system).

Parameters influencing powder rolling are:

(i) Roll gap: As the roll gap is increased, strip thickness also increases. This leads to a decrease in green density of the strip. On the other hand, use of a very small roll gap leads to edge cracking.

(ii) Roll diameter: This is closely related to strip thickness as well as green density of the strip. Also, for a given strip thickness, as the roll diameter increases, both density and green strength will increase.

(iii) Roll speed: Speed of the rolls is usually kept low, about 0.3 to 0.5 m/s.

(iv) Powder characteristics: Irregular powders with rough surfaces provide better interlocking and hence higher strip density

- A relatively new development in powder rolling is the use of slurry instead of powders.
- In the slurry method, powders are mixed with suitable binders to form slurry. This slurry is used to prepare

Sintering

- The main purpose of sintering in powder rolling is to improve the properties
- In addition, sintering also helps in the removal of undesirable elements such as oxygen and Sulphur.
- During sintering in a hydrogen atmosphere, oxide films present on the surface are removed.
- In the case of nickel powders produced by hydrometallurgical technique, sintering serves to remove Sulphur through the formation of hydrogen sulphide.
- Cracked ammonia, endogas or vacuum can also be used for sintering.
- Use of protective atmosphere during sintering also helps to avoid the oxidation of surface and subsurface pores.
- Sintering is carried out using a continuous belt furnace and the sintered strip passes on to the rolling

Densification of sintered strip

- Sintered strip is porous and for achieving full density, it must be re-rolled.
- Densification of the sintered strip is aimed at removing the residual porosity (~10%).
- This densification step is not done for applications where porosity is desired.
- Densification is achieved either

(i) by repeated cold rolling followed by annealing

(i) In the repeated cold work-annealing method, several cycles may be needed before the required density is achieved.

This will depend on the amount of porosity present in the sintered material.

The amount of cold work that can be given to the sintered strip increases with increasing density of the strip.

In this mode of densification, it is essential to match the rate of production of green strip with that sintering rate and rate of rolling so that the process is continuous.

(ii) Hot Rolling: Hot sintered strip is rolled

Final cold rolling and annealing

- Hot rolling is followed by a separate final cold rolling and annealing step.

- For strips densified by repeated cold rolling and annealing, the two steps (rolling and annealing) can be combined.

Advantages of Powder Rolling

- Powder rolled sheets have **no crystallographic orientation**
- Powder rolling does not require large capital investment.
- A metal strip can be clad on one or both the sides by using it as a carrier of the selected powder through the rolls.
- A larger percentage of any scrap material reclaimed in powder form can be reused.
- The process may be made continuous by passing the green strip through a sintering furnace and then to re-rolling.
- Strips exhibiting properties equivalent to those of wrought material can be successfully produced by powder rolling.
- In comparison to conventional casting and rolling, powder rolling gives a rolled strip free from defects such as laminations.
- The process is also economic since conventional route involves fettling, annealing and pickling of the rolled product resulting in a low yield.
- **This is particularly important in cases where cheap powders with high work hardening characteristics are to be processed.**

Limitations

- The leakage of powders sideways during rolling is a major problem. This leakage causes ends of the strip to be fragile.
- Variation in density. Density variation across the length of the strip can be avoided by proper cleaning of roll surface between successive rolling operations.
- Density variation across the width is a result of powder leakage causing edges to have lower densities than other regions.
- This difficulty can be overcome by (i) using a sealant (a continuous belt that seals the gap at the roll edges, or (ii) by having undercuts in the rolls or (iii) by adding floating

- Need for large diameter rolls for producing thin strips as compared to conventional rolling.
- This is due to the fact that the starting material is a powder.
- The angle at which the powders are gripped by the rolls is, therefore, different from the calculated values because of the slipping of powders.
- **Roll diameters must be up to 100 times or more than the strip thickness desired.**

Influence of Powder Characteristics

- The particle size of the powders has a definite influence on the thickness and quality of the strip produced.
- The powder used must be fairly coarse, or else leakage of powders through the rolls will occur.
- For thin strips and foils, fine powders may be used, while heavy sheet requires coarse powders with a normal particle size distribution.
- Higher the initial density and more the reduction in thickness of the initial powder feed, greater will be the final density.
- Particle characteristics like density as well as its compactibility control the final strip density.
- Irregular shape powders give higher reduction in

Applications

- Nickel strips for coinage
- Cobalt powder is processed by powder rolling to yield strips.
- Multilayer strips for use in bearings. These strips contain a stainless-steel backing strip with Cu-Pb-Sn layer over it. Another example is the rolling of Al-Pb-Sn multilayer strips.
- Nickel-iron strips for controlled expansion properties.
- Cu-Ni-Sn alloys for electronic applications.
- Porous nickel strip for alkaline batteries and